digital

USER MANUAL

PDP-11/05 PANEL REAWAKENING

Transformed into an Interactive Multi-Media Art Piece



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Table of Contents

Table of Contents 3
Overview 6
Hardware Components 6
  Interface Board 6
  Control Board 6
  Power and Mounting 6
Functional Overview 8
  Modes 8
Switch Groupings 9
  Address/Data Switches 9
  Sense Switches 9
Audio Visualizations 10
  Music Styles 10
  Voice Styles 10
  Where Each Style Applies 10
Switch Echo Mode (0 / 000) 11
Binary Clock Mode (1 / 001) 12
  Displaying the Time 12
  Setting the Time 12
  Temperature Display 12
  Timer 13
Blinkenlights Computer Mode (2 / 010) 14
  Audio 14
  Display Effects 16
  Favorites 16
Themes Soundboard Mode (3 / 011) 17
  Themes 17
  Playback 17
Knight Rider Mode (4 / 100) 19
Jukebox Mode (5 / 101) 20
  Folder Selection 20
  Track Selection 20
  Controls 20
Bluetooth Audio Mode (6 / 110) 21

Flash Settings Mode (7 / 111) 22
Address Select 22
Provisioning Reset 22
Editing a Value 22
Reset 22
VU Calibration 22
Settings Table 23
Power-On Greeting 24
Panel Lock: Self-Destruct Sequence 26
Appendix I: Jukebox Channels 27
Channel #1 – Electronic Ambient Covers 27
Channel #2 – Ambient Electronic I 27
Channel #3 – Ambient Electronic II 28
Channel #4 – Synthwave 28
Channel #5 – Artemis Station 28
Channel #6 – Cadwell's Reach 29
Channel #7 – Echelon Protocol 29
Channel #8 – Navigating a Latent Space 29
Channel #9 – Arrival at Myriah's Shadow 29
Channel #10 – Night City 29
Channel #11 – Nova Sonus 29
Channel #12 – To a Distant World 30
Channel #13 – Dune (Full Album) 30
Channel #14 – Computer Love (Remaster) 30
Channel #15 – Computer World (Full Album) 30
Channel #16 – The Colours of Infinity 30
Channel #17 – Pink Floyd Style Ambient Music 31
Channel #18 – Voyage 34 (Full Album) 32
Channel #19 – As Falls Wichita, So Falls Wichita Falls (Full Album) 32
Channel #20 – I Dream of Star Trek 32
Channel #21 – Forbidden Planet Electronic Tonalities 32
Channel #22 – Do it Yourself 32
Channel #23 – Initial call to the Military Fire Station from Chernobyl 33
Channel #24 – Jupiter II Engines 33
Channel #25 – Theme from "The Twilight Zone" 33

Channel #26 – Theme from “The Outer Limits” 33
Channel #27 – Calibration Tone 33

Overview

This project centers on a Digital Equipment Corporation PDP-11/05 front control panel — the switch-and-lamp interface once used to examine, deposit, halt, and step through memory on the original minicomputer. The panel is being brought back to working, lit condition and reframed as a multi-media art piece, pairing restored vintage hardware with modern electronics to give the panel a second life beyond its original computing role. This project represents the culmination of two years of spare time effort.

This document serves as the user manual for the piece — covering what it is, how to operate and interact with it, and what to expect from it. A separate document covers the technical design and implementation details.

Hardware Components

The electronics consist of an interface board, a control board, and a power supply. Each board is double-sided.

Interface Board

The interface board reads the panel's switches and drives its LED indicators. It performs the following functions:

- Reads switch states via PISO (parallel-in, serial-out) shift registers
- Drives the LED display via SIPO (serial-in, parallel-out) shift registers combined with a 4x16 multiplexer
- Bridges the panel's legacy TTL-level signaling using modern CMOS HCT-family logic
- Reads the panel lock key switch input
- Performs level shifting between logic domains

Control Board

The control board consists of the following:

- ESP32 microcontroller
- BT201 sound card with SD card and Bluetooth capability
- DS3231 real-time clock
- Analog conditioning and protection between the BT201 and the ESP32 ADC input

Audio output from the BT201's 3.5mm headphone jack is routed by cable to a powered speaker. A Bose SoundLink Mini is provided for this purpose.

Power and Mounting

A 5V, 2A wall supply provides more than adequate power for the panel and its electronics. The panel and electronics are mounted on a repurposed Apple Cinema Display stand. It's the grandchild supporting the grandparent. For scale, the ESP32 controlling the panel runs over a thousand times faster than the original 11/05, with several times the RAM of a fully loaded 11/05 and dozens of times its capacity in flash.



The completed PDP-11/05 Panel Reawakening.



Rear view, showing the back plate mounted to the stand.



The stand, with the added DEC logo on its base.

Functional Overview

Modes

The panel's mode switches (Low3) select one of eight modes. Toggling START selects the mode set on the Low3 switches:

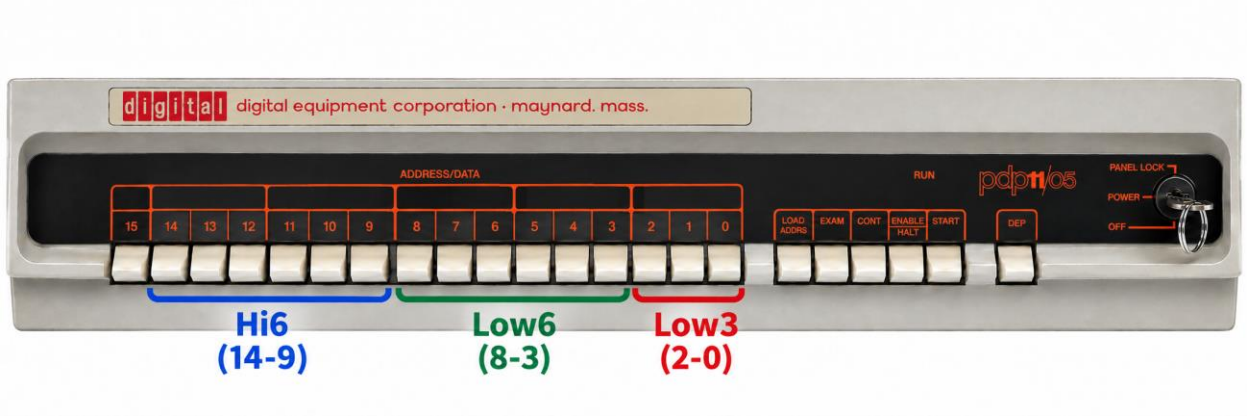
- 0 (000) – Switch echo to LED display
- 1 (001) – Binary clock with Westminster chime.
  Sub modes include:
  - Time setting function
  - Temperature indication
  - Presettable countdown timer
- 2 (010) – "Blinkenlights Computer" and audio computer effects
- 3 (011) – Soundboard
  - 0 (000) – Random theme cycling (without replacement)
  - 1 (001) – Robby the Robot, from Forbidden Planet
  - 2 (010) – The B9 Environmental Robot, from Lost in Space
  - 3 (011) – The HAL 9000 computer, from 2001: A Space Odyssey
  - 4 (100) – Sound effects from the original Star Trek series
  - 5 (101) – Sound effects from Star Wars
  - 6 (110) – Sound effects from Colossus: The Forbin Project
  - 7 or greater – Chaos: a random clip from a random theme, chosen anew for every clip
- 4 (100) – Homage to the Knight Rider series
- 5 (101) – Jukebox music player (from SD card)
- 6 (110) – Bluetooth audio receiver
- 7 (111) – Flash update functions

Throughout the audio-driven modes, the LED display comes alive alongside the sound, offering a variety of visualizations that range from classic VU meter animations to voice displays that dance with the dialogue and effects.

- At power on, a random or pre-selected verbal greeting is played.
- In addition to the eight switch-selected modes, placing the key switch in the panel lock position triggers a separate feature: a 10 minute animated self-destruct sequence.
- The blue LED on the esp32 flashes every second to indicate that the controller is "alive".

Switch Groupings

The panel's toggle switches and sense switches are referred to by name throughout the mode descriptions that follow.



Address/Data Switches

The 16 address/data toggle switches are divided into three named groups:

- Hi6 – switches 14 through 9
- Low6 – switches 8 through 3
- Low3 – switches 2 through 0

Switch 15, the leftmost, belongs to no group and is not used for mode selection or settings. It is used only as a display bit in the binary clock and as part of the full 16-bit switch code in Flash Settings Mode.

Sense Switches

To the right of the address/data switches are the panel's sense switches: LOAD ADDRS, EXAM, CONT, ENABLE/HALT, START, and DEP. All sense switches other than ENABLE/HALT are momentary, spring-loaded switches, pressed downward and then released, except DEP, which is pressed upward and then released. ENABLE/HALT is a toggle switch that works identically to the address/data switches.

Audio Visualizations

Several modes drive the LED display in sync with whatever audio is playing, using one of a shared set of visualization styles. Pressing EXAM cycles through the styles available in the current context. The styles fall into two families: music styles and voice styles.

Music Styles

- L2RVU (default) – a traditional VU bar that fills left to right.
- R2LVU – the mirror image, filling right to left.
- Center – radiates outward from the center.
- Edge – radiates inward from both edges.
- Binary – displays the audio level as a binary count, climbing from the least significant bit to the most.

Voice Styles

- VOXC – radiates outward from the center in response to speech.
- VOXE – radiates inward from the edges in response to speech.
- B9 – flashes the entire display on and off with speech, an homage to the B9 robot from Lost in Space.

Where Each Style Applies

- The power-on greeting and the Soundboard mode offer only the voice styles.
- The Jukebox mode offers only the music styles.
- The Bluetooth audio receiver mode offers both families.

Switch Echo Mode (0 / 000)

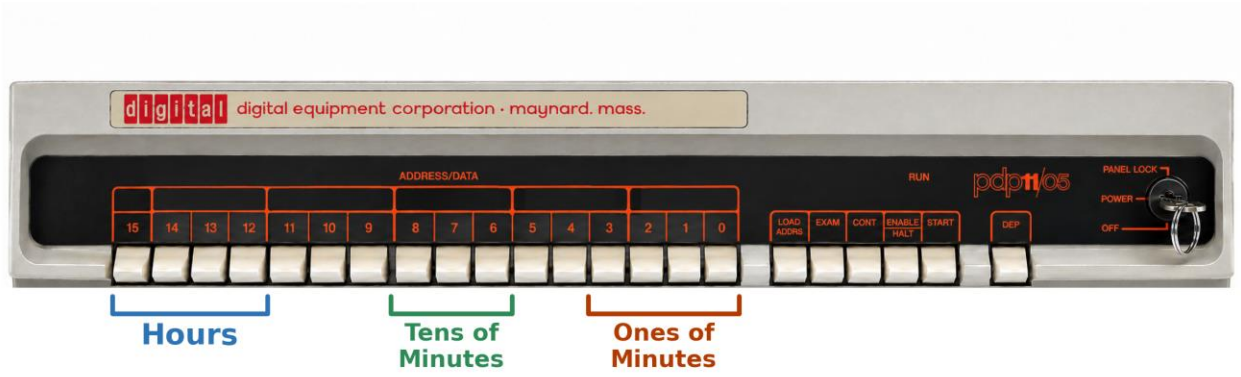
With Low3 set to 0 (000) and START toggled, the address/data switches are echoed directly on the LED display: each active switch lights its corresponding LED.

This mode is useful as a demonstration, a test of the switches and display, or simply as a numeric reminder.

Binary Clock Mode (1 / 001)

With Low3 set to 1 (001) and START toggled, the panel operates as a binary clock, displaying the time of day on the LED array with an optional Westminster chime and a configurable tick volume. The CONT sense switch cycles through the clock's three modes: time, temperature, and a countdown timer.

The time is displayed in a highly readable BCD-style format: hours occupy bits 15 through 12, tens of minutes occupy bits 8 through 6, and ones of minutes occupy bits 3 through 0. Note that the ones-of-minutes field overlaps both Low3 and the lowest bit of Low6.



Displaying the Time

- The current time is shown on the LED display.
- On the quarter hour, if a chime is selected, a Westminster chime plays (bim-bam and hour-strike are Westminster variants).
- Between chimes, if the tick volume is set above off, a tick sounds each second at the selected volume.
- All clock audio, including the chime, is muted whenever ENABLE/HALT is engaged.
- Pressing EXAM while the time is displayed triggers a chime test, playing a one-time preview of the 3:00 chime. Unlike normal chime playback, the chime test is not affected by ENABLE/HALT.

Setting the Time

- Press LOAD while the time is displayed to enter time-setting mode.
- The display mirrors the address/data switches live as the desired hour and minute are entered.
- The RUN light turns on once a valid time is set.
- Press DEP to commit the new time; the RUN light turns off and a confirmation tone plays.

Temperature Display

- Reached by pressing CONT from the time display.
- Shows the approximate ambient temperature in Fahrenheit, as measured by the clock module's built-in sensor.

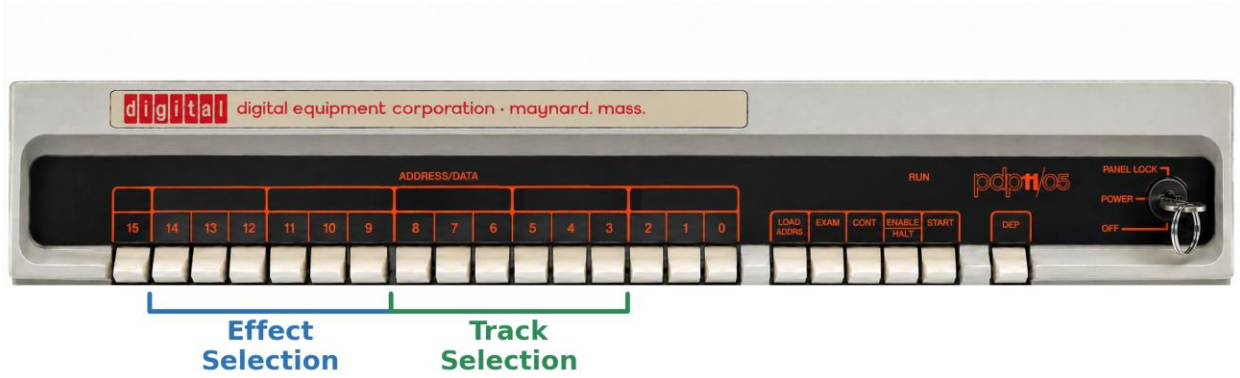
- This is a rough estimate. It reflects the chip's own temperature, can take a while to settle, and can be calibrated if needed.

Timer

- Reached by pressing CONT again, from the temperature display.
- Set the desired countdown duration using the switches, then press DEP to start the countdown.
- If the tick volume is set above off, a tick sounds each second at the selected volume, unless ENABLE/HALT is engaged.
- When the countdown reaches zero, an alarm sounds. Afterward, the clock returns to the timer setting screen, ready to be reused.
- Pressing CONT at any point while in the Timer mode returns directly to the time display.

Blinkenlights Computer Mode (2 / 010)

With Low3 set to 2 (010) and START toggled, the panel drives a synchronized pairing of audio and animated LED display effects, evoking a busy vintage-computer aesthetic. Audio and display are selected independently and run independently of one another.



Audio

- By default, tracks from the sound library play in a shuffled rotation.
- Setting Low6 to a value from 1 to 19 repeats that specific track continuously instead. Any other value falls back to shuffled playback.
- LOAD pauses and resumes playback; DEP fast-forwards.
- CONT skips to the next track in shuffled mode, or restarts the current track from the beginning in repeat mode.
- Engaging ENABLE/HALT lets the current playback finish before stopping: with a specific track selected, playback stops once that track completes; in shuffled mode, it stops once the full rotation of tracks has cycled through.

Audio tracks:

Low6	Track
1	PDP-11/40 (Source: Bill T. Degnan)
2	PDP-11/20
3	Daisy
4	Alien's Mother Computer
5	10 Minutes of Ambient Computer
6	Colossus
7	Retrofuturistic Computer
8	80's Computer Interface
9	Star Trek Computer #1
10	Star Trek Computer #2
11	Star Trek Computer #3
12	Star Trek Computer #4

Low6	Track
13	Star Trek Type 7 Computer
14	Eminiar 7 Battle Computer
15	Star Trek Various TNG Beeps
16	Library Computer On Bridge
17	Computer Room
18	Wargames Effect
19	WOPR Ambience Sound

Display Effects

- By default, the LED display automatically cycles through a set of animated effects, each running for a configurable interval before moving to the next.
- Setting Hi6 to a value from 1 to 24 selects one of the effects directly, listed below. The change takes effect immediately, no need to press START again.
- Returning Hi6 to 0 resumes automatic cycling.

Available display effects:

Hi6	Effect
1	Panel
2	Binary Counter
3	Up/Down Counter
4	Criss Cross
5	Random
6	Checkerboard
7	Wig Wag
8	Diamond
9	Dissipate
10	Chaser
11	Random Bit Fill
12	Snake
13	Split
14	Join
15	Implode
16	Implode II
17	Squeeze
18	Marquee
19	Chopsticks
20	Octal Count
21	Octal Fill
22	Pulse
23	Idle Computer
24	Knight Rider

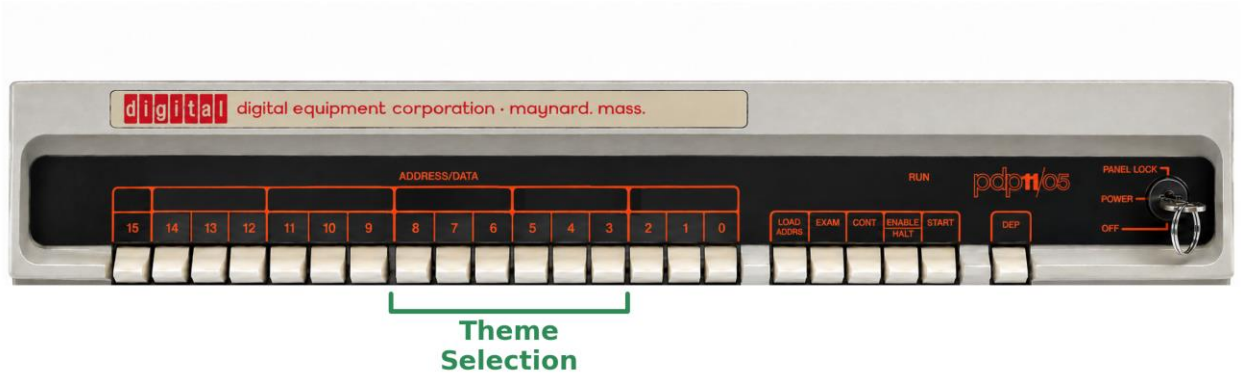
Favorites

Some audio and display combinations work especially well together:

Low6	Audio	Hi6	Effect
1	PDP-11/40	1	Panel
3	Daisy	19	Chopsticks

Themes Soundboard Mode (3 / 011)

With Low3 set to 3 (011) and START toggled, the panel plays themed sound clips on a timer, with a voice-reactive VU meter animation on the LED display synchronized to whichever clip is playing.



Themes

- Setting Low6 to a value from 1 to 6 pins the soundboard to one specific theme, listed below.
- Setting Low6 to 0 enables random theme cycling (without replacement): once every clip in the current theme has played, a new theme is chosen at random.
- Setting Low6 to 7 or greater enables chaos mode: a new random theme is chosen for every clip, and a random clip is played from it.
- The Low6 setting is re-checked before each clip, so it can be changed at any time and takes effect at the next clip, without restarting the mode.

Available themes:

Low6	Theme
1	Robby the Robot (Forbidden Planet)
2	The B9 Environmental Robot (Lost in Space)
3	The HAL 9000 Computer (2001: A Space Odyssey)
4	Star Trek (original series)
5	Star Wars
6	Colossus (The Forbin Project)

Playback

- In fixed-theme and random-theme modes, clips within a theme play in random (without replacement) order until every clip has been heard. In random-theme mode, the next theme then begins.

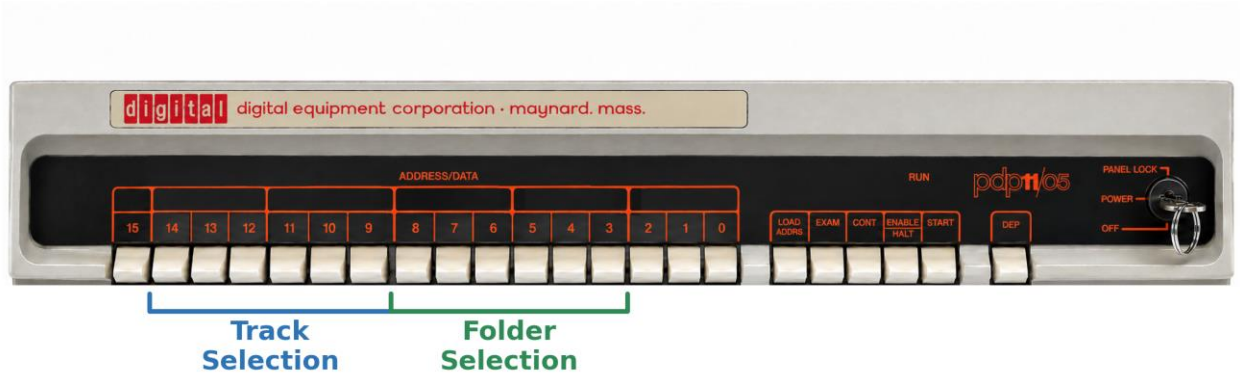
- In chaos mode, every clip is picked independently, so a clip can repeat before others have played. The VU meter style follows the style of each clip's theme, unless changed with EXAM.
- A pause separates each clip from the next. The pause length is the sum of a fixed interval and a random interval, both intervals are flash-configurable (see Flash Settings Mode).
- Between clips, the LED display may run a quiet idle animation, a back-and-forth sentry sweep or a random flickering dot, until the next clip begins.
- Pressing EXAM cycles through the available voice-style VU meter displays (see Audio Visualizations).
- Engaging ENABLE/HALT pauses the soundboard. A clip that is already playing finishes, but no further clip starts while ENABLE/HALT is engaged. Once ENABLE/HALT is released, the next clip plays after the normal pause between clips.

Knight Rider Mode (4 / 100)

With Low3 set to 4 (100) and START toggled, the panel pays homage to Knight Rider. A single audio track plays alongside the Knight Rider scanning-light display effect (see the Display Effects table, entry 24). Audio and display start together and stop together, with no theme or track selection involved. Engaging ENABLE/HALT stops the sequence from automatically repeating.

Jukebox Mode (5 / 101)

With Low3 set to 5 (101) and START toggled, the panel becomes a music jukebox, playing folders of tracks from the SD card with an accompanying VU meter animation. The full folder and track listing is provided in Appendix I.



Folder Selection

- Setting Low6 to a valid folder number selects that folder (see Appendix I for the full list).
- An invalid folder number turns on the RUN light and no track plays.
- In shuffle mode, the folder and track selections are re-read at the end of each cycle, so switching between cycles doesn't interrupt play already in progress. In sequential mode, once started, the folder plays to the end and loops back to track 1, without re-checking the switches; changing folders or tracks takes effect only on the next restart of the mode.

Track Selection

- Setting Hi6 to a valid track number for the current folder drops the needle at that track and plays sequentially to the end of the folder, then loops back to track 1.
- Setting Hi6 to anything other than a valid track shuffles the whole folder instead. Once every track has played once, a new random cycle begins.
- Once a shuffle cycle is underway, Hi6 isn't re-read until the cycle finishes, preserving uninterrupted, LP-style play.

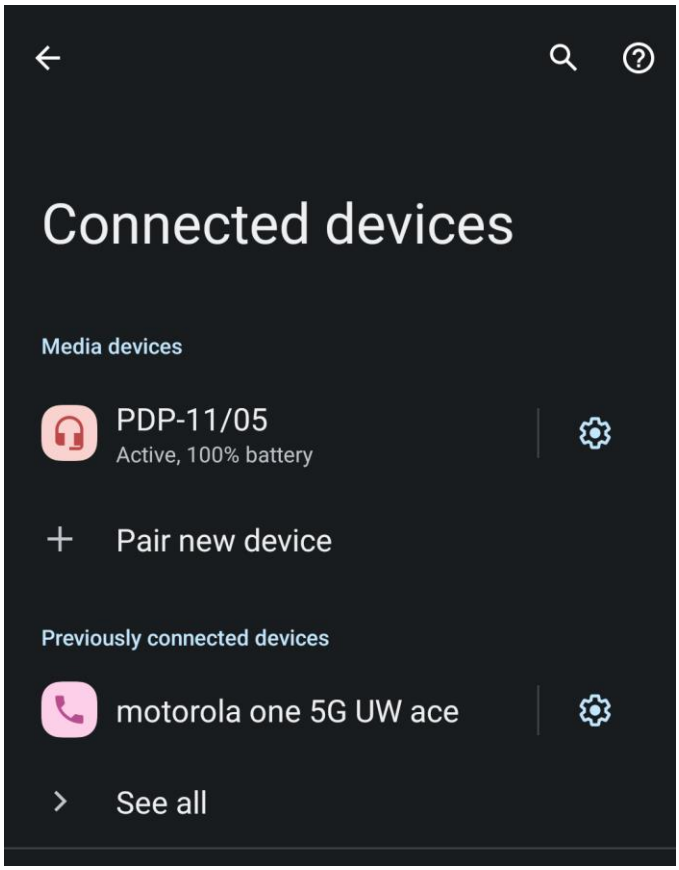
Controls

- LOAD pauses and resumes playback.
- DEP fast-forwards the current track; with ENABLE/HALT also engaged, it rewinds instead.
- CONT skips ahead to the next track.
- EXAM cycles through the available music-style VU meter displays (see Audio Visualizations). The chosen style carries over between tracks, but resets to the default on restart.
- ENABLE/HALT acts as a “stop when done” flag: the current sequence or shuffle cycle is allowed to finish naturally, then playback stops rather than looping. A CONT press overrides this for one track, though ENABLE/HALT reasserts at the next cycle boundary.

Bluetooth Audio Mode (6 / 110)

With Low3 set to 6 (110) and START toggled, the panel acts as a Bluetooth audio receiver, pairing with a nearby device and displaying its audio on the LED VU meter.

- The panel's Bluetooth device name is PDP-11/05; look for that name when pairing or checking your phone's connected devices list.
- The Bluetooth source is typically a cell phone.
- Volume can be controlled from the phone. Keep the phone's volume reasonable to avoid overdriving the sound level.
- Pressing EXAM cycles through all available VU meter styles, both music and voice (see Audio Visualizations).
- Leaving this mode returns the panel to its normal audio mode, restoring the prior VU meter settings and, depending on configuration, the normal playback volume.



The panel appears as "PDP-11/05" under Media devices on the phone's Connected devices screen.

Flash Settings Mode (7 / 111)

With Low3 set to 7 (111) and START toggled, the panel exposes an interface for viewing and adjusting the persistent settings stored in the system's flash memory. Pressing CONT cycles through three top-level modes: Address Select, Reset, and VU Calibration.

Address Select

- The address/data switches directly select a settings-table address, shown live on the LED display.
- The RUN light turns on when the address is valid (see the settings table below).
- Press LOAD to open that setting's value for editing.

Provisioning Reset

- From Address Select, keying in switch code 0177570 (octal) and pressing CONT triggers a BT201 provisioning reset, restoring the Bluetooth module's factory settings. A confirmation tone plays, and the panel remains in Address Select.
- A power cycle is required afterward for the change to take effect. On the next power-up, the system detects that the Bluetooth device name has changed and reconfigures the module accordingly, after which a second power cycle is required.

Editing a Value

- Reached from Address Select via LOAD.
- The current stored value is shown; setting the address/data switches replaces it with the new value to be written.
- The RUN light turns on when the switches represent a value within that setting's valid range.
- Press DEP to write the new value to flash (a confirmation tone plays), returning to Address Select. Press CONT instead to leave without committing, moving on to Reset.

Reset

- Displays an octal 052525 (alternating bits) prompt.
- Press DEP to restore every setting to its factory default (a confirmation tone plays); remains in Reset. Press CONT to move on to VU Calibration.

VU Calibration

- Displays the current DC offset used by the VU meter's audio sampling.
- Press DEP to re-measure and save a new offset (a confirmation tone plays on success); remains in VU Calibration. Press CONT to loop back to Address Select.

Settings Table

The following settings are available at their corresponding address:

Key: User setting, Advanced user setting, Developer setting

Addr	Name	Default	Range	Description
1	tick	1	0–3	Clock tick volume: 0 = off, 1 = low, 2 = medium, 3 = high
2	chime	1	0–3	Clock chime: 0 = off, 1 = Westminster, 2 = Westminster bim-bam variant, 3 = Westminster hour-strike variant
3	cycleTime	30	10–1800	Seconds between display animation changes for Blinkenlights effect when random effect is selected.
4	tempCal	8	0–32	Onboard temperature calibration offset in degrees Fahrenheit (16 = 0, 0 = -16, 32 = +16)
5	fixIntrvl	15	0–65535	Fixed interval for timed soundboard events, in seconds
6	rndIntrvl	15	0–65535	Random additional interval for soundboard events, in seconds
7	idlPattrn	1	0–2	Idle animation pattern selector: 0 = off, 1 = sentry, 2 = random dot
8	introTrack	0	0–15	Power-on greeting track (0 = random). See power on greeting section below.
9	introStyle	1	0–3	Power-on greeting VU meter style (0 = random, 1–3 = specific)
10	ovrBarCnt	10	0–16	Override countdown bar count
11	fnlBarCnt	10	0–16	Final countdown bar count
12	peakDecay	150	0–1000	VU meter peak hold decay rate
13	VuGain	64	1–100	SD card playback VU meter gain
14	BinGain	2048	1–65535	SD card playback Binary-style VU meter gain (multiplier; larger = more sensitive)
15	btVuGain	39	1–100	Bluetooth VU meter gain
16	btBinGain	128	1–65535	Bluetooth Binary-style VU meter gain (multiplier; larger = more sensitive)
17	plRunLt	1	0–1	Panel lock RUN light effect: 1 = enabled, 0 = disabled
18	plReboot	30	0–3600	Reboot delay after the self-destruct sequence, in seconds (0 = no reboot)
19	plVol	25	1–30	Panel lock playback volume
20	plAudThrsh	12	1–200	Panel lock audio detection threshold
21	plLagMs	1750	0–5000	Panel lock audio detection lag, in milliseconds
22	plNumCross	3	0–10	Panel lock consecutive-crossing noise filter count
23	volume	16	1–30	Normal playback volume
24	stepCount	20	1–300	Fast-forward/rewind step count (each step = 0.2s; 300 = 1 minute)
25	debounce	8	1–32	Switch debounce time, in milliseconds

Addr	Name	Default	Range	Description
26	btVolReset	1	0–1	1 = restore normal volume level when exiting Bluetooth mode
27	dcOffset	1978	0–4095	ADC baseline offset for the VU meter
28	settleTime	1	1–8	Switch settle time, in microseconds
29	vuAttack	31	0–65535	L2RVU/R2LVU meter attack time constant, in 0.1 ms units (31 = 3.1 ms). Larger = slower, 0 = instant
30	vuDecay	1990	0–65535	L2RVU/R2LVU meter decay time constant, in 0.1 ms units (1990 = 199 ms). Larger = slower, 0 = instant
31	ctrAttack	90	0–65535	Center/Edge meter attack time constant, in 0.1 ms units (90 = 9 ms)
32	ctrDecay	390	0–65535	Center/Edge meter decay time constant, in 0.1 ms units (390 = 39 ms)
33	b9Attack	190	0–65535	B9 attack time constant, in 0.1 ms units (190 = 19 ms)
34	b9Decay	990	0–65535	B9 decay time constant, in 0.1 ms units (990 = 99 ms)
35	binAttack	4000	0–65535	Binary meter attack time constant, in 10 ms units (4000 = 40 s). A slow attack makes the display climb like a counter while audio plays
36	binDecay	2490	0–65535	Binary meter decay time constant, in 0.1 ms units (2490 = 249 ms)
37	voxAttack	90	0–65535	VOXC/VOXE attack time constant, in 0.1 ms units (90 = 9 ms)
38	voxDecay	390	0–65535	VOXC/VOXE decay time constant, in 0.1 ms units (390 = 39 ms)

The VU attack and decay settings (addresses 29–38) are advanced settings that control how quickly the VU meter display responds to the audio level:

- Each value is a time constant: larger values respond more slowly, and 0 responds instantly. Units are 0.1 ms (31 = 3.1 ms), except binAttack, which is in 10 ms units (4000 = 40 s).
- Attack sets how quickly the display rises with the audio level; decay sets how quickly it falls.
- Each pair applies to a group of styles: vu to L2RVU and R2LVU, ctr to Center and Edge, vox to VOXC and VOXE, b9 to B9, and bin to Binary.
- BinGain and btBinGain multiply the audio level for the Binary style. With the slow binAttack default, the Binary display counts upward while audio plays, and the gain sets how fast it counts.
- Changes take effect the next time the VU meter starts.

Power-On Greeting

At power-up, before any mode is selected, the panel plays a short greeting with a voice-reactive VU meter animation (see Audio Visualizations).

- The introTrack setting selects a specific greeting from the table below. A value of 0 plays a random greeting instead.
- The introStyle setting selects a specific voice VU meter style. A value of 0 selects one at random.
- Once the greeting finishes, the panel proceeds to normal operation.

Available greetings:

introTrack	Source	Greeting
1	HAL 9000	"I feel much better now, I really do."
2	AI	Intro to PDP-11/05 front panel
3	Lost in Space (B9 Robot)	"This project is extremely dangerous."
4	HAL 9000	"Everything is going extremely well."
5	HAL 9000	"Good afternoon gentlemen, I am a HAL 9000 computer."
6	HAL 9000	"Good evening Dave."
7	Short Circuit	"Need input"
8	Short Circuit	"Nice software!"
9	War Games	"Shall we play a game?"
10	Star Trek	"Working"
11	War Games	"Greetings professor Falken"
12	AOL	"You've got mail."
13	Battlestar Galactica	"By your command"
14	Lost in Space (B9 Robot)	"Thank you"
15	Knight Rider	KITT introduction

Panel Lock: Self-Destruct Sequence

Turning the key switch to the PANEL LOCK position triggers a self-destruct sequence: a narrated countdown to destruction, accompanied by animated LED effects synchronized to the audio. This is separate from the eight switch-selected modes and can be triggered regardless of the current mode switch setting.

- The display animations follow the narration, shifting between radiating patterns, voice-reactive displays during announcements, countdown displays, and finally a destruction sequence.
- Early in the sequence, the narration offers an option to override. Returning the key switch to its normal position during this window aborts the sequence and returns the panel to whatever mode was previously selected.
- Once that window has expired, the sequence commits permanently: the RUN light begins flashing, and returning the key switch will no longer stop it.
- After the sequence completes, the panel either restarts on its own or holds dark until power-cycled, depending on how it is configured.
- Best experienced rather than described. Turn the key and watch it play out.

Appendix I: Jukebox Channels

The full list of Jukebox channels (folders), selected via Low6, and the tracks within each.

Channel #1 – Electronic Ambient Covers

Electronic ambient covers performed by State Azure:

Track #	Track Name	Artist
1	In the Air Tonight	State Azure
2	Love comes quickly	State Azure
3	Save a prayer	State Azure
4	Running up that Hill	State Azure
5	Rachel's Song	State Azure

Channel #2 – Ambient Electronic I

Ambient electronic tracks by State Azure

Track #	Track Name	Artist
1	Paradise Star	State Azure
2	Immaculate Sky	State Azure
3	Hereafter	State Azure
4	Shades	State Azure
5	Cave of the Crystal Harrier	State Azure
6	The Tesseract	State Azure
7	Sanctuary	State Azure
8	Invisible Light	State Azure
9	Canis Majoris	State Azure
10	Alpha Core	State Azure
11	Pale Oasis	State Azure
12	Mass Effect	State Azure
13	Perspective	State Azure
14	Xenogenesis	State Azure
15	Passing Storm	State Azure

Channel #3 – Ambient Electronic II

Ambient electronic tracks by various artists

Track #	Track Name	Artist
1	Love On A Real Train	Tangerine Dream
2	Monday Paracetamol	Ulrich Schnauss
3	Here Today, Gone Tomorrow	Ulrich Schnauss
4	Mothers of Rain	Tangerine Dream
5	King Ghost	Steven Wilson & Tangerine Dream
6	Matjora is Still Alive	Johannes Schmoelling
7	Mosaic	Paul Haslinger
8	Vanishing Blue	Tangerine Dream
9	Eternity	Stellardrone
10	Etheric Echoes	Dreamstate Logic
11	Remote Viewing	Tangerine Dream
12	Morgenstern	Schiller & Quasching
13	Dem Himmel So Nah	Schiller & Quasching
14	Midsommer	Schiller & TQ

Channel #4 – Synthwave

1980s Synthwave, Miami Vice style.

Track #	Track Name	Artist
1	Hard Evidence	Sun City
2	Crockett's Theme	Jan Hammer
3	City Limits	State Azure
4	Rain	Jan Hammer
5	Switchblade	State Azure
6	Homage to Jan Hammer	State Azure
7	Airwolf Theme	Sylvester Levay (State Azure cover)

Channel #5 – Artemis Station

4+ Hour Semi-Generative, Meditative Ambient

Track #	Track Name	Artist
1	Artemis Station	State Azure

Channel #6 – Cadwell's Reach

4 Hour Ambient

Track #	Track Name	Artist
1	Cadwell's Reach	State Azure

Channel #7 – Echelon Protocol

1 Hour Ambient Mix

Track #	Track Name	Artist
1	Echelon Protocol	State Azure

Channel #8 – Navigating a Latent Space

Track #	Track Name	Artist
1	Navigating a Latent Space	State Azure

Channel #9 – Arrival at Myriah's Shadow

1.5 Hour Sci-Fi Ambient

Track #	Track Name	Artist
1	Arrival at Myriah's Shadow	State Azure

Channel #10 – Night City

4 Hour Cyberpunk Ambient Mix

Track #	Track Name	Artist
1	Night City	State Azure

Channel #11 – Nova Sonus

2 Hour Ambient Mix

Track #	Track Name	Artist
1	Nova Sonus	State Azure

Channel #12 – To a Distant World

Electronic Ambient

Track #	Track Name	Artist
1	Part 1	State Azure
2	Part 2	State Azure
3	Part 3	State Azure
4	Part 4	State Azure

Channel #13 – Dune (Full Album)

"Klaus Schulze's 1979 album inspired by Frank Herbert's novel"

Track #	Track Name	Artist
1	Dune	Klaus Schulze
2	Shadows of Ignorance	Klaus Schulze
3	Le Mans	Klaus Schulze

Channel #14 – Computer Love (Remaster)

2009 Remaster

Track #	Track Name	Artist
1	Computer Love	Kraftwerk

Channel #15 – Computer World (Full Album)

Kraftwerk's 1981 album themed around computer technology

Track #	Track Name	Artist
1	Computer World ("Computerwelt")	Kraftwerk
2	Pocket Calculator ("Taschenrechner")	Kraftwerk
3	Numbers ("Nummern")	Kraftwerk
4	Computer World 2 ("Computerwelt 2")	Kraftwerk
5	Computer Love ("Computer Liebe")	Kraftwerk
6	Home Computer ("Heimcomputer")	Kraftwerk
7	"It's More Fun to Compute"	Kraftwerk

Channel #16 – The Colours of Infinity

Unreleased music by David Gilmour from the 1995 documentary featuring Arthur C. Clarke

Track #	Track Name	Artist
1	Infinity	David Gilmour

Channel #17 – Pink Floyd Style Ambient Music

2+ Hour AI generated ambient mix inspired by the music of Pink Floyd

(Source: Echoes Sound Lab)

Track #	Track Name	Artist
1	Track #1	PFAMBIEN
2	Track #2	PFAMBIEN
3	Track #3	PFAMBIEN
4	Track #4	PFAMBIEN
5	Track #5	PFAMBIEN
6	Track #6	PFAMBIEN
7	Track #7	PFAMBIEN
8	Track #8	PFAMBIEN
9	Track #9	PFAMBIEN
10	Track #10	PFAMBIEN
11	Track #11	PFAMBIEN
12	Track #12	PFAMBIEN
13	Track #13	PFAMBIEN
14	Track #14	PFAMBIEN
15	Track #15	PFAMBIEN
16	Track #16	PFAMBIEN
17	Track #17	PFAMBIEN
18	Track #18	PFAMBIEN
19	Track #19	PFAMBIEN
20	Track #20	PFAMBIEN
21	Track #21	PFAMBIEN
22	Track #22	PFAMBIEN
23	Track #23	PFAMBIEN
24	Track #24	PFAMBIEN
25	Track #25	PFAMBIEN
26	Track #26	PFAMBIEN
27	Track #27	PFAMBIEN
28	Track #28	PFAMBIEN
29	Track #29	PFAMBIEN

Channel #18 – Voyage 34 (Full Album)

A psychedelic musical journey

Track #	Track Name	Artist
1	Phase I	Porcupine Tree
2	Phase II	Porcupine Tree
3	Phase III	Porcupine Tree
4	Phase IV	Porcupine Tree

Channel #19 – As Falls Wichita, So Falls Wichita Falls (Full Album)

Contemporary Jazz (fusion)

Track #	Track Name	Artist
1	As Falls Wichita, So Falls Wichita Falls	Pat Metheny and Lyle Mays
2	Ozark	Pat Metheny and Lyle Mays
3	September 15th (dedicated to Bill Evans)	Pat Metheny and Lyle Mays
4	It's for You	Pat Metheny and Lyle Mays
5	Estupenda Graça	Pat Metheny and Lyle Mays

Channel #20 – I Dream of Star Trek

A hard to find mix tape featuring electronic music with Star Trek overdubs

Track #	Track Name	Artist
1	I Dream of Star Trek	Various artists

Channel #21 – Forbidden Planet Electronic Tonalities

The groundbreaking electronic music from the movie "Forbidden Planet"

Track #	Track Name	Artist
1	Soundtrack	Louis & Bebe Barron
2	Mixed Emotions (2000)	Bebe Barron

Channel #22 – Do it Yourself

You can build a mainframe from the things you find at home!

Track #	Track Name	Artist
1	Do it Yourself	Bill Sutton

Channel #23 – Initial call to the Military Fire Station from Chernobyl

Recording of the first call to the Military Fire Station right after the Chernobyl explosion.

Track #	Track Name	Artist
1	Chernobyl Fire Station Call	N/A

Channel #24 – Jupiter II Engines

The sound of the Jupiter II deutronium-annihilation drive engines from "Lost in Space"

Track #	Track Name	Artist
1	J2 Engines	Ralph Hickey

Channel #25 – Theme from "The Twilight Zone"

Theme music from the TV show "The Twilight Zone"

Track #	Track Name	Artist
1	"The Twilight Zone" theme music	Marius Constant

Channel #26 – Theme from "The Outer Limits"

Theme music from the TV show "The Outer Limits"

Track #	Track Name	Artist
1	"The Outer Limits" opening theme	Dominic Frontiere

Channel #27 – Calibration Tone

1 kHz test tone at 50% modulation, useful for calibrating SD card playback volume when the volume parameter changes

Track #	Track Name	Artist
1	1 kHz Test Tone (50% Modulation)	N/A